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(54) **STATOR HAVING A WIRE WITH A
SLACKENED PORTION**

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See application file for complete search history.

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Primary Examiner — Jose A Gonzalez Quinones

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H02K 15/32 (2025.01)

H02K 24/00 (2006.01)

(52) **U.S. Cl.**

CPC **H02K 3/522** (2013.01); **H02K 15/32**
(2025.01); **H02K 24/00** (2013.01)

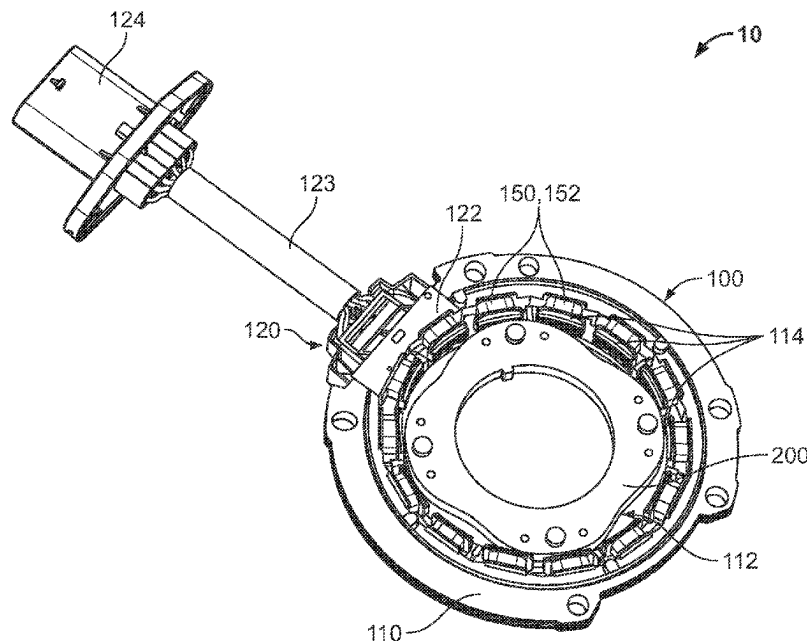
(58) **Field of Classification Search**

CPC H02K 3/522; H02K 15/32; H02K 24/00;
G01D 5/20

(57) **ABSTRACT**

A stator of a resolver includes a body having a pole, a connector interface disposed in a portion of the body and including a terminal having a wire termination at a termination end, and a wire disposed around the pole and extending to the termination end of the terminal. The terminal has a bent section at the termination end that is bent to a bending angle greater than 0° and less than or equal to 90° with respect to a main section of the terminal. The wire is electrically and mechanically connected with the terminal at the wire termination. The bent section of the terminal forms a slackened portion of the wire adjacent to the wire termination.

17 Claims, 9 Drawing Sheets



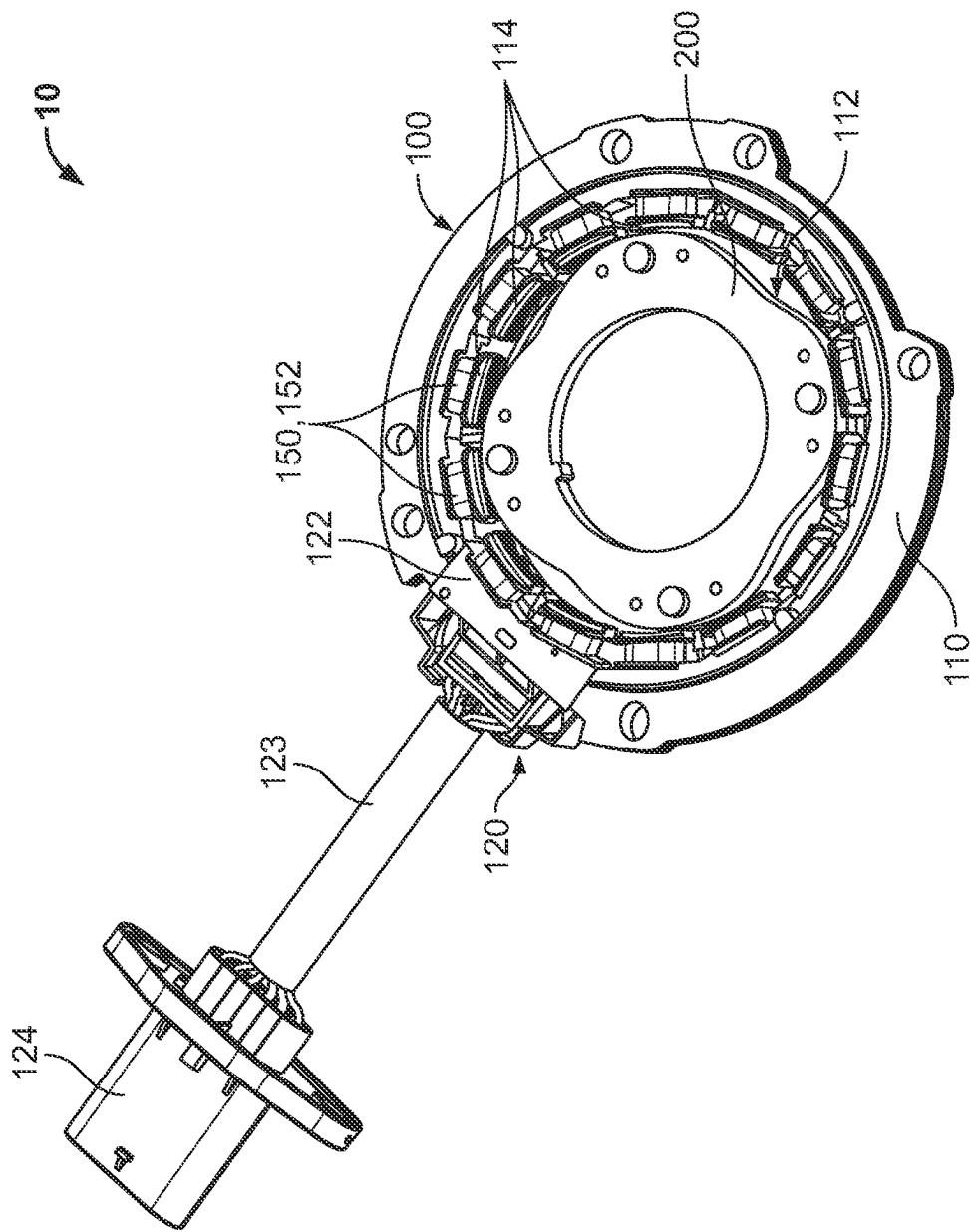


Fig. 1

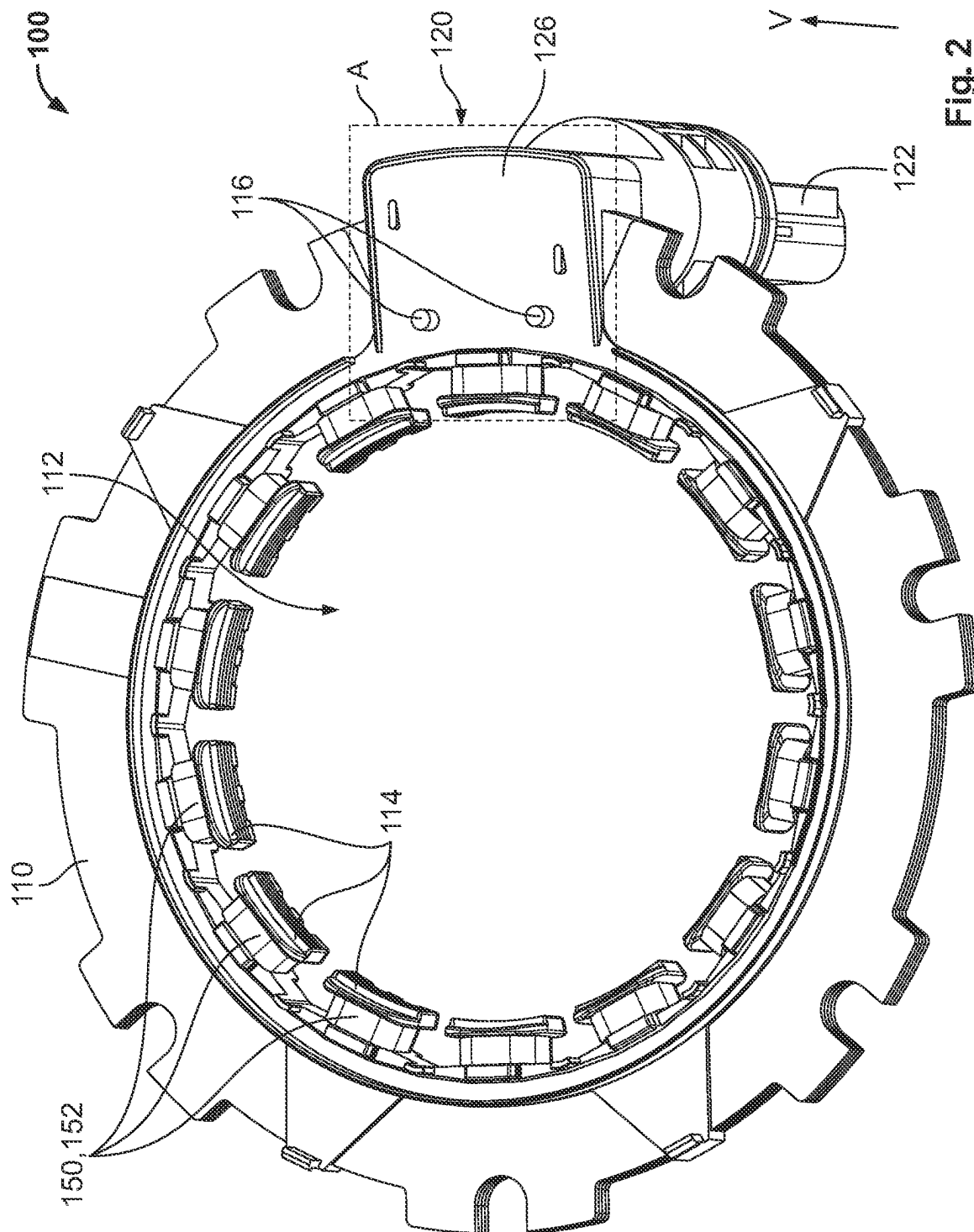
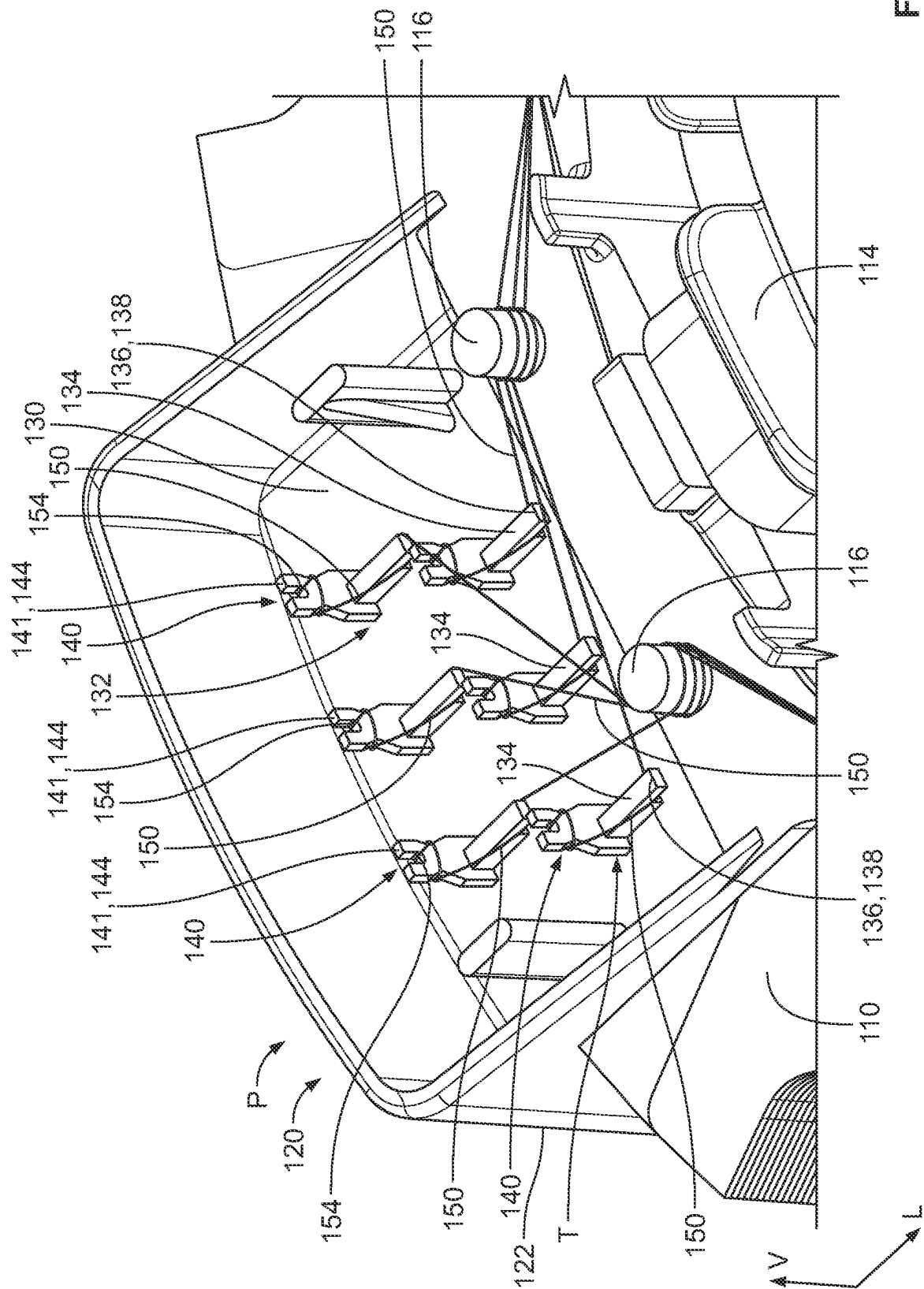


Fig. 2



3
5
4

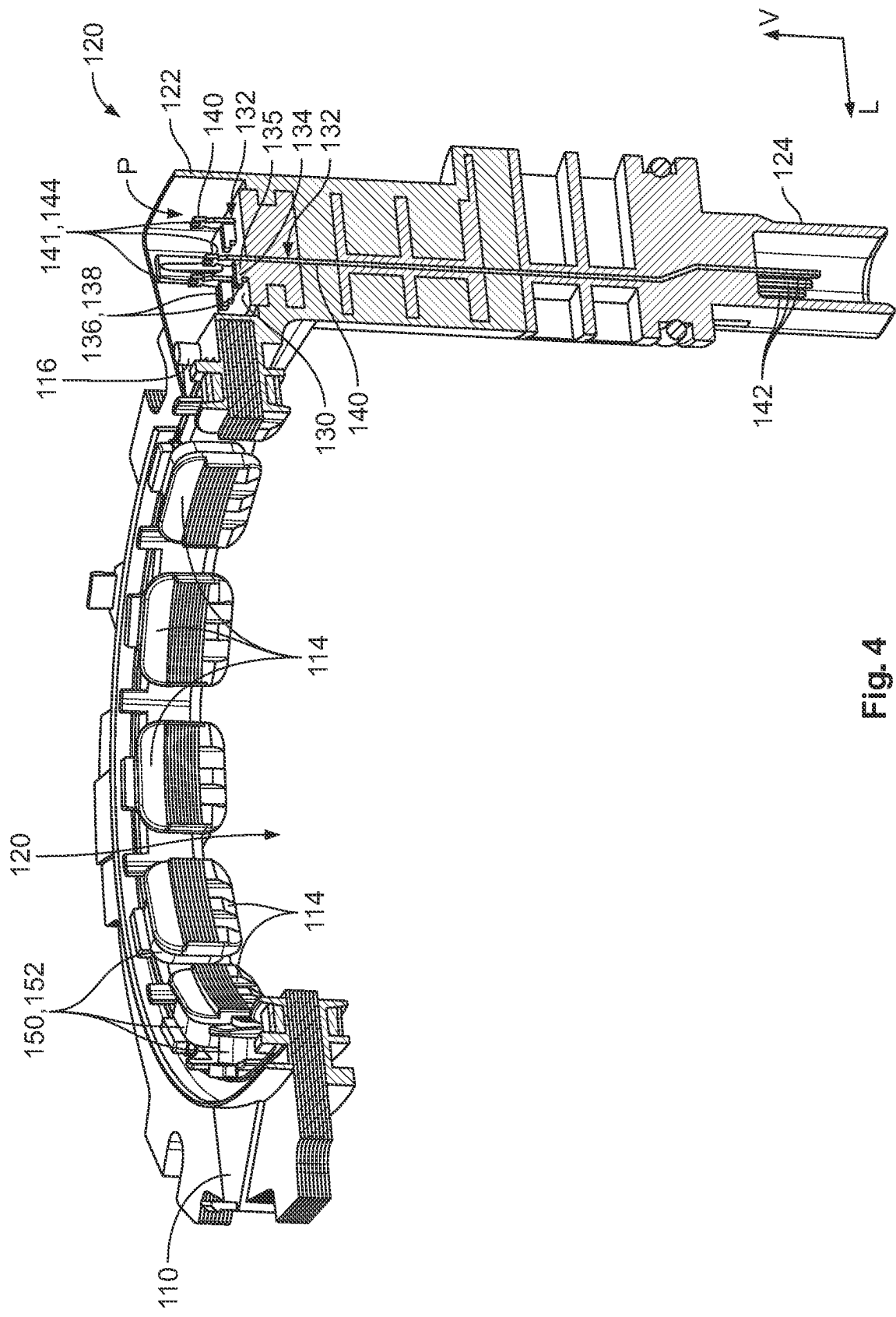


Fig. 4

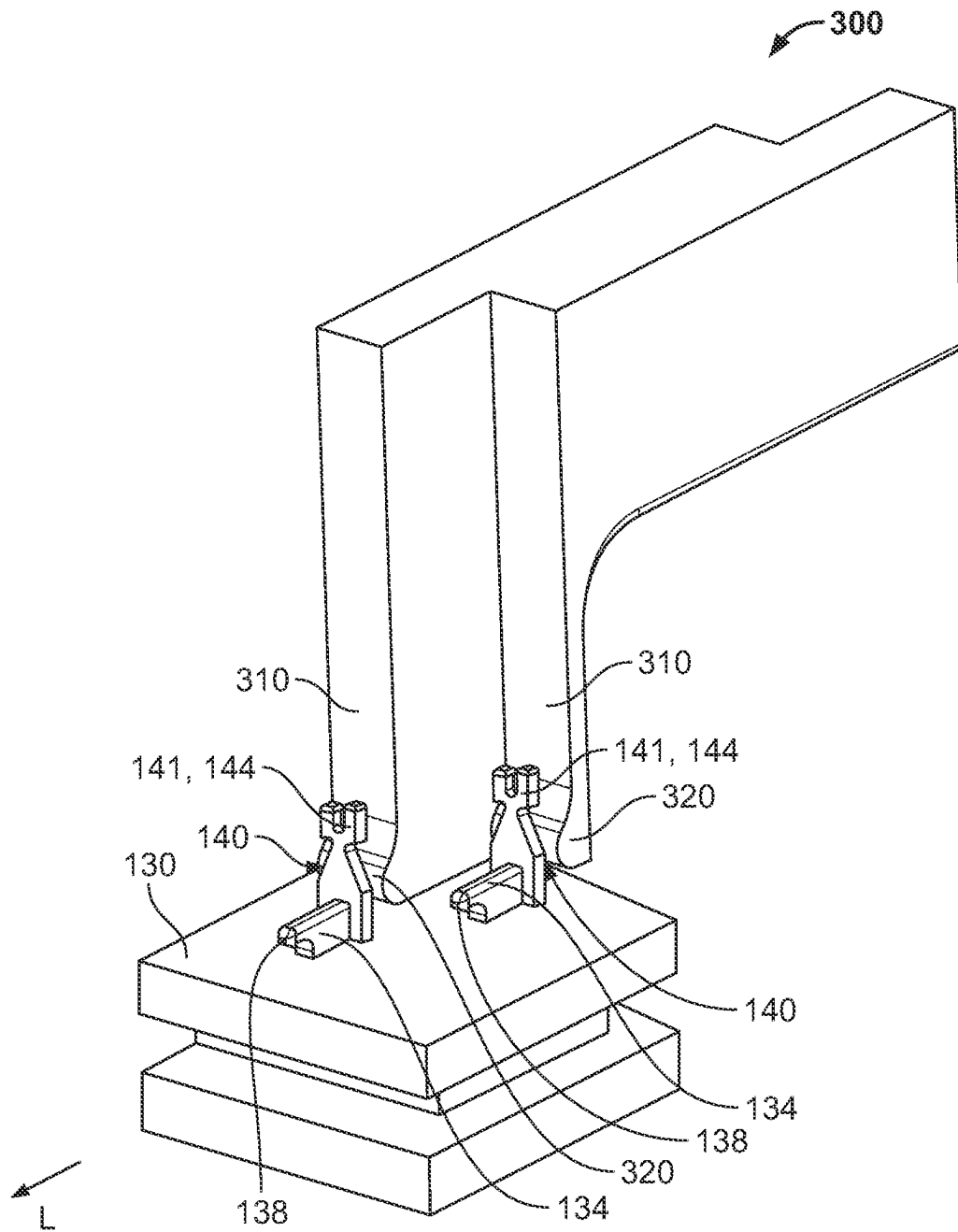


Fig. 5

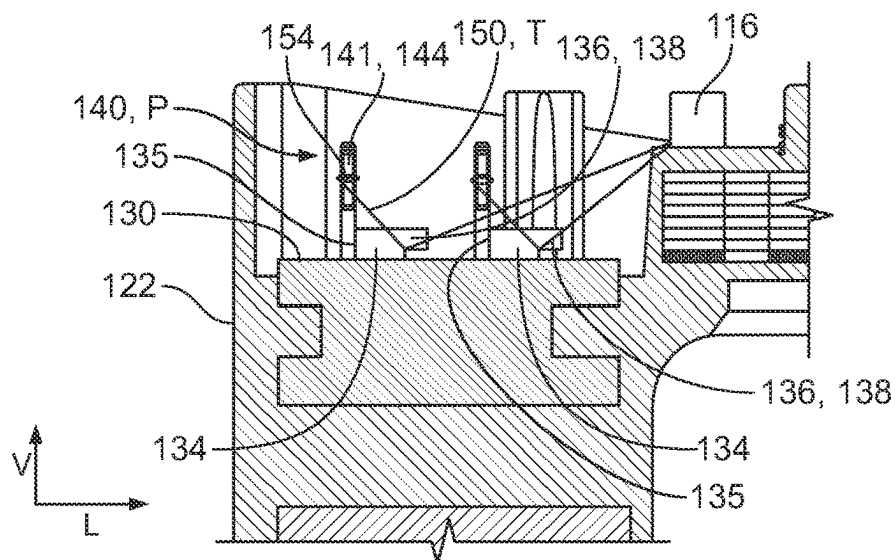


Fig. 6A

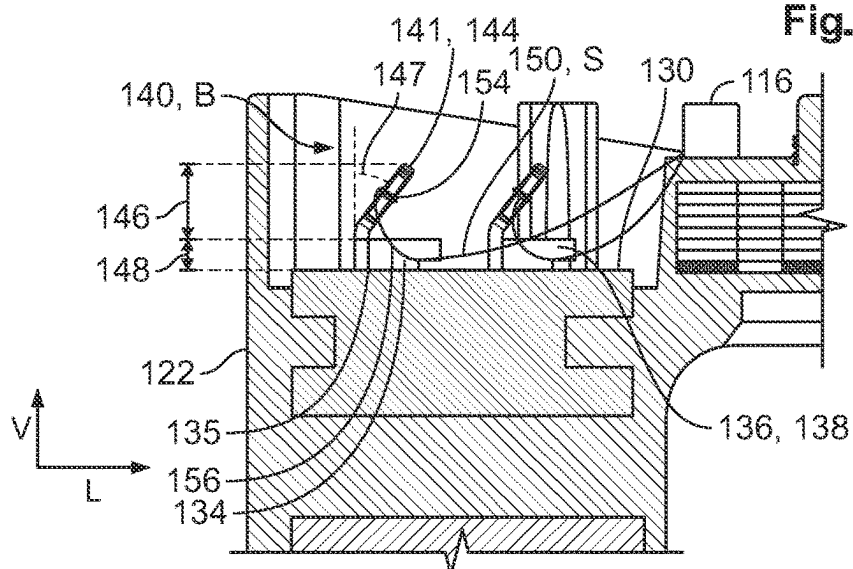


Fig. 6B

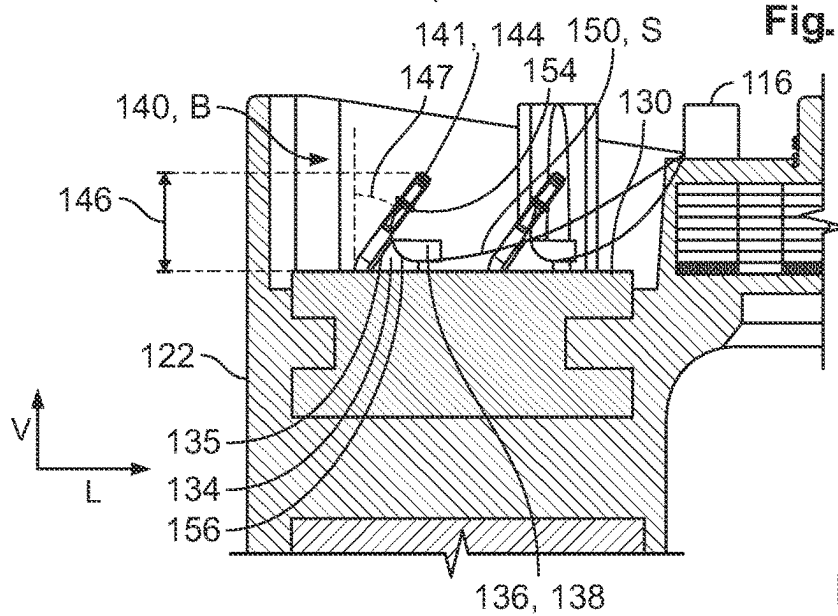


Fig. 6C

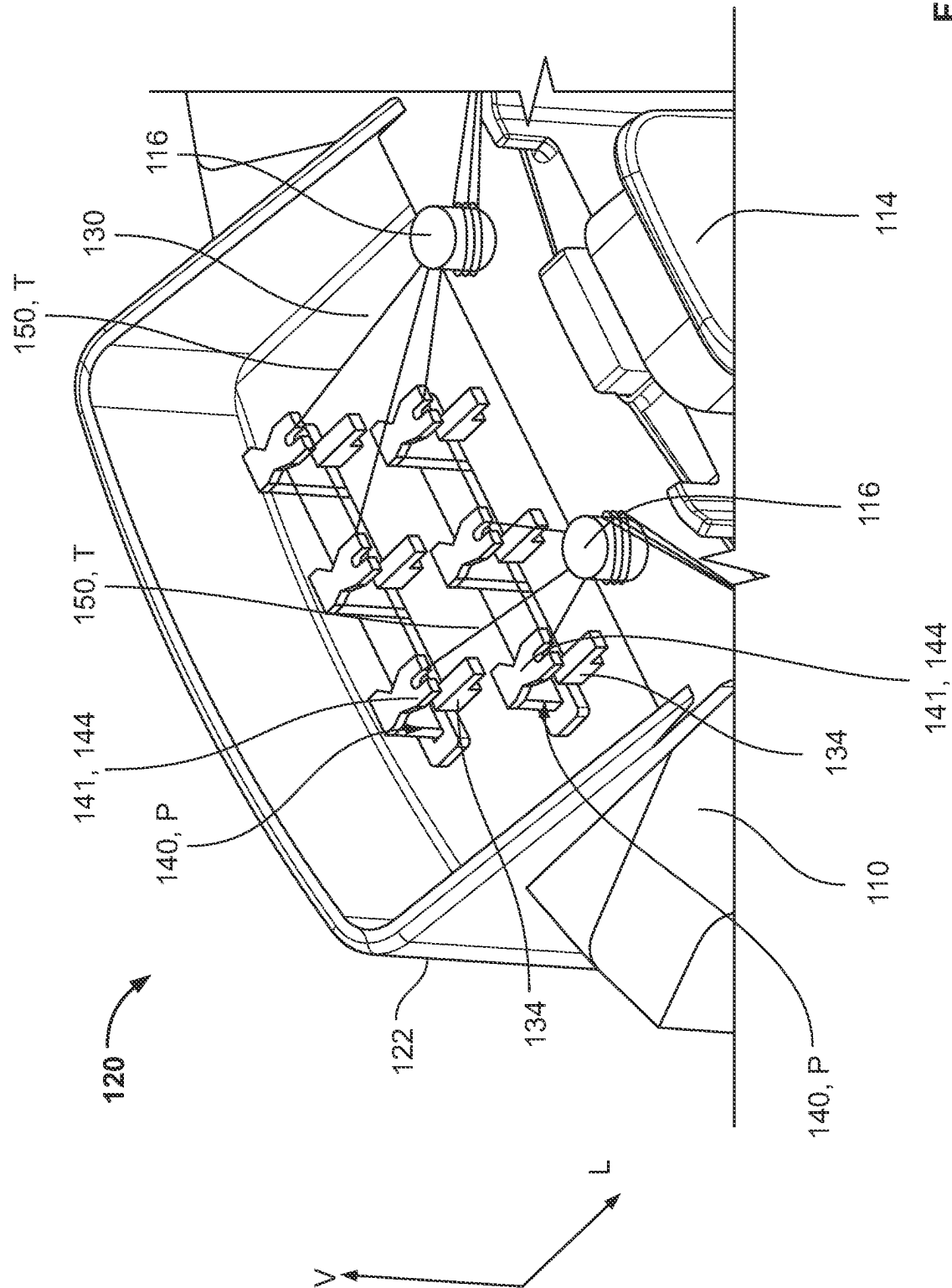


Fig. 7A

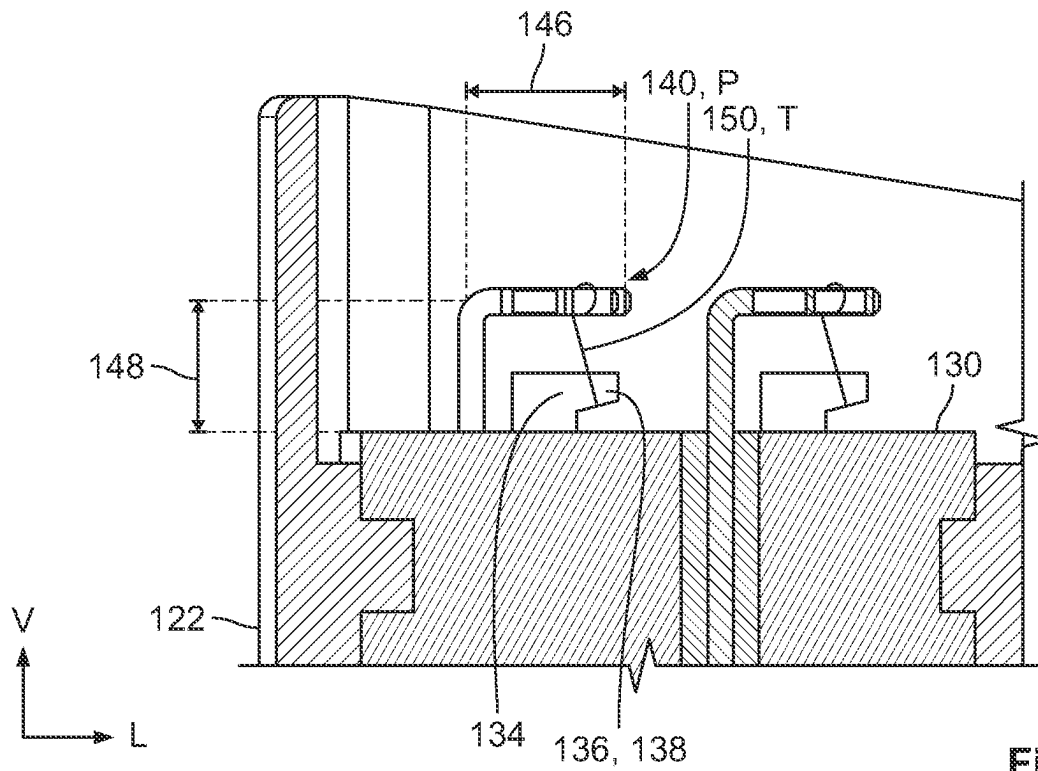


Fig. 7B

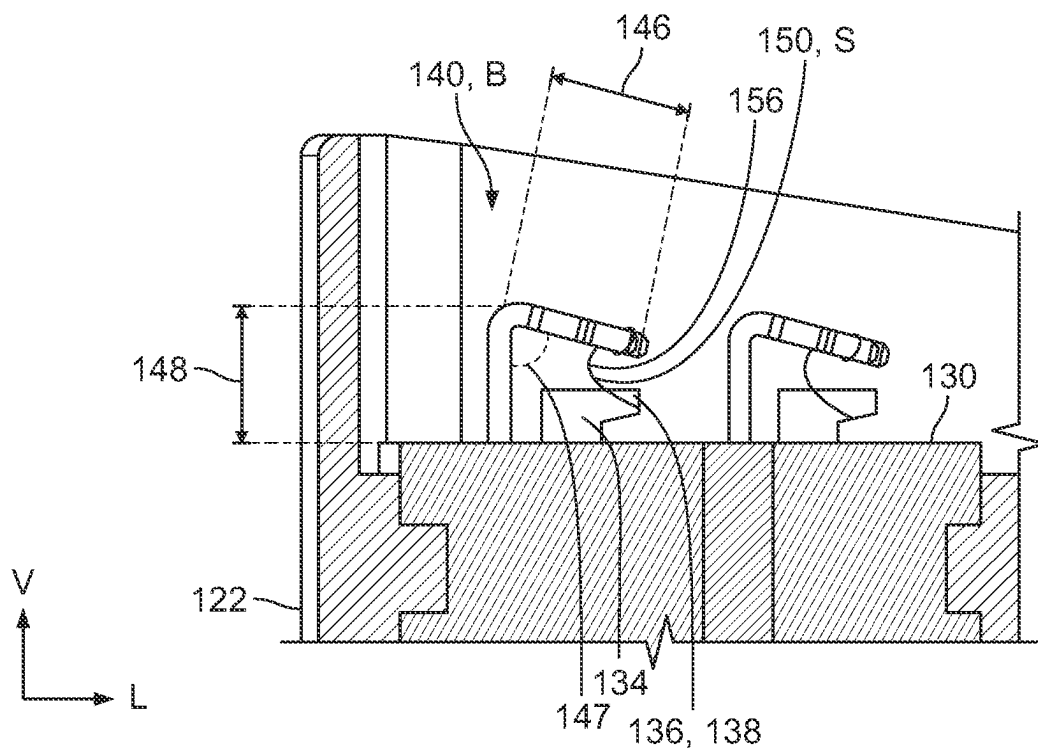
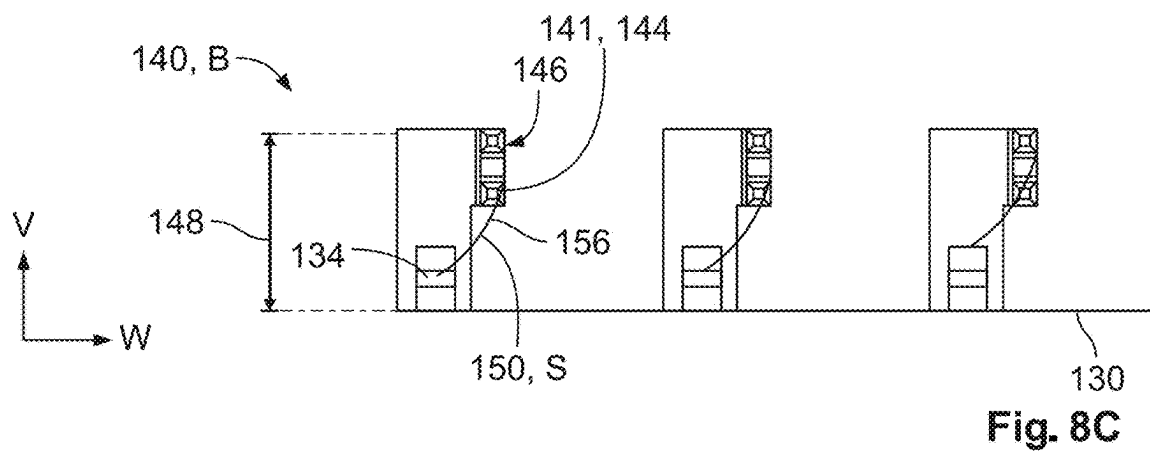
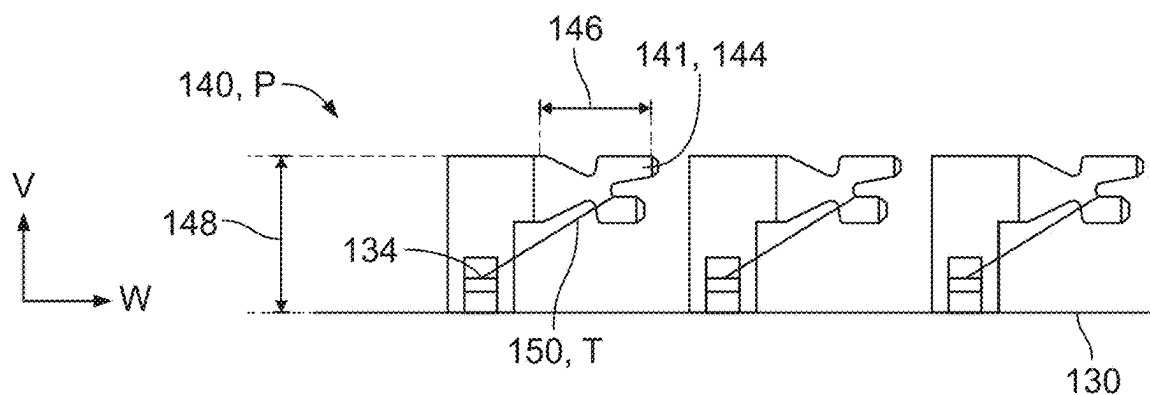
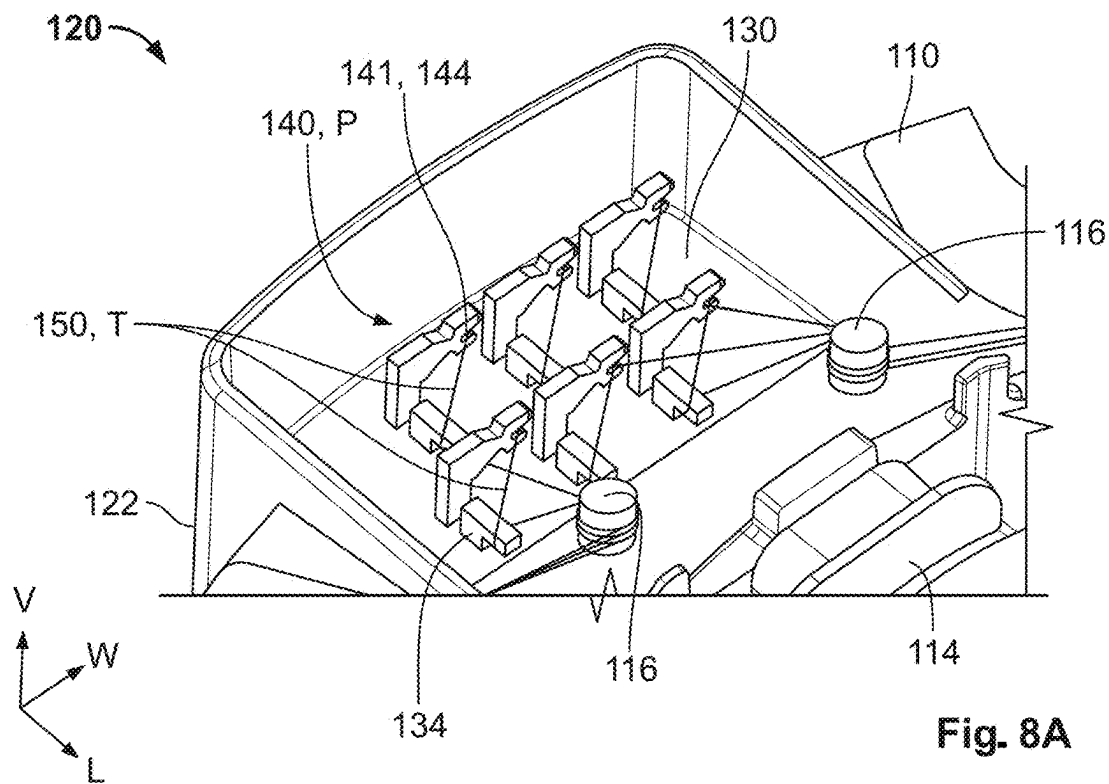


Fig. 7C



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STATOR HAVING A WIRE WITH A SLACKENED PORTION

FIELD OF THE INVENTION

The present invention relates to a stator of a resolver and, more particularly, to tension of a wire of the stator.

BACKGROUND

A resolver generally includes a stator and rotor disposed within the stator. The stator determines a rotational position of the rotor using wires wound around a plurality of poles that face the rotor. Each of the wires extends from one of the poles to a terminal of the stator. The wire is in tension and generally taut when it is connected to the terminal.

Resolvers can be subject to harsh environments in various applications. Rapid temperature change, for example, in these harsh environments causes strain on the tensioned wire that can result in breakage of the wire.

Some resolvers provide strain relief to the wires of the stator in order to avoid damage to or breakage of the wire under harsh conditions. These resolvers commonly include an external element that is placed on the stator and is in contact with the wires when the wires are connected in tension to the terminals; the external element is then removed to allow for slack in the wires that can compensate for temperature change. These solutions, however, require additional process steps and are time consuming to implement, while also requiring additional components that increase the cost of material and the cost of production. Further, removing the external element can cause damage to the wires.

SUMMARY

A stator of a resolver includes a body having a pole, a connector interface disposed in a portion of the body and including a terminal having a wire termination at a termination end, and a wire disposed around the pole and extending to the termination end of the terminal. The terminal has a bent section at the termination end that is bent to a bending angle greater than 0° and less than or equal to 90° with respect to a main section of the terminal. The wire is electrically and mechanically connected with the terminal at the wire termination. The bent section of the terminal forms a slackened portion of the wire adjacent to the wire termination.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention will now be described by way of example with reference to the accompanying Figures, of which:

FIG. 1 is a perspective view of a resolver according to an embodiment;

FIG. 2 is a perspective view of a stator of a resolver according to an embodiment;

FIG. 3 is a detail perspective view of a portion A of FIG. 2 without a cover;

FIG. 4 is a sectional perspective view of the stator of FIG. 2;

FIG. 5 is a perspective view of a bending tool according to an embodiment with terminals and a terminal base of a stator;

FIG. 6A is a sectional side view of a connector interface of the stator of FIG. 2 with terminals in a pre-bent state;

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FIG. 6B is a sectional side view of the connector interface of FIG. 6A with the terminals in a bent state;

FIG. 6C is a sectional side view of a connector interface with terminals according to another embodiment in a bent state;

FIG. 7A is a perspective view of a connector interface having terminals according to another embodiment in a pre-bent state;

FIG. 7B is a sectional side view of the connector interface of FIG. 7A with the terminals in the pre-bent state;

FIG. 7C is a sectional side view of the connector interface of FIG. 7B with the terminals in the bent state;

FIG. 8A is a perspective view of a connector interface having terminals according to another embodiment in a pre-bent state;

FIG. 8B is a schematic side view of the connector interface of FIG. 8A with the terminals in the pre-bent state; and

FIG. 8C is a schematic side view of the connector interface of FIG. 8B with the terminals in the bent state.

DETAILED DESCRIPTION OF THE EMBODIMENT(S)

Exemplary embodiments of the present disclosure will be described hereinafter in detail with reference to the attached drawings, wherein like reference numerals refer to like elements. The present disclosure may however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, these embodiments are provided so that the present disclosure will convey the concept of the disclosure to those skilled in the art. In addition, in the following detailed description, for purposes of explanation, numerous specific details are set forth to provide a thorough understanding of the disclosed embodiments. However, it is apparent that one or more embodiments may also be implemented without these specific details.

Throughout the drawings, only one of a plurality of identical elements may be labeled in a figure for clarity of the drawings, but the detailed description of the element herein applies equally to each of the identically appearing elements in the figure. Throughout the specification, directional descriptors are used such as “vertical”, “longitudinal”, and “width”. These descriptors are merely for clarity of the description and for differentiation of the various directions. These directional descriptors do not imply or require any particular orientation of the disclosed elements.

A resolver **10** according to an embodiment is shown in FIG. 1. The resolver **10** includes a stator **100** and a rotor **200** disposed within the stator **100**. The rotor **200** is rotatable with respect to the stator **100** and the stator **100** has an output indicating a rotational position and/or a rotation speed of the rotor **200**.

The stator **100** is shown in multiple embodiments in FIGS. 1 and 2. In each embodiment, the stator **100** includes a body **110**, a connector interface **120** disposed in a portion of the body **110**, and a plurality of wires **150** extending from the body **110** to the connector interface **120**.

As shown in FIGS. 1 and 2, the body **110** is an approximately circular member defining a rotor receiving space **112**. The rotor **200** is disposed in the rotor receiving space **112**. The body **110** has a plurality of poles **114** positioned around a circumference of the rotor receiving space **112** and facing into the rotor receiving space **112**. The poles **114** are positioned at a regular interval around the circumference and are spaced apart from one another. The body **110** has a pair

of posts **116**, shown in FIGS. **2** and **3**, extending in a vertical direction **V** and positioned adjacent to or within the connector interface **120**.

The connector interface **120** is shown generally in the embodiments of FIGS. **1** and **2**, and a detail view of a portion **A** of FIG. **2** including the connector interface **120** is shown in FIG. **3**. The connector interface **120** includes a connector housing **122**, a terminal base **130** disposed within the connector housing **122**, and a plurality of terminals **140** disposed within the terminal base **130**.

In the embodiment of FIG. **1**, the connector housing **122** is disposed on an edge of the body **110** and the connector interface **120** has a wire **123** extending between the connector housing **122** and a mating portion **124** of the connector interface **120**. In another embodiment shown in FIGS. **2** and **4**, the connector housing **122** is an elongated member extending in the vertical direction **V** perpendicular to a plane of the body **110**; in this embodiment, the connector housing **122** extends continuously to the mating portion **124** and no wire is positioned between the connector housing **122** and the mating portion **124**, as shown in FIG. **4**. The connector interface **120** will be explained in detail with reference to the embodiment shown in FIGS. **2-4**, however, in other embodiments, the connector housing **122** could alternatively extend in other directions to the mating portion **124**, including to the left, right, or flat in a plane of the body **110**. In all embodiments of the connector interface **120**, the features of the terminal base **130** and the terminals **140** are the same as described herein.

As shown in FIG. **2**, the connector housing **122** includes a cover **126** covering an end of the connector housing **122** in the vertical direction **V**. The connector housing **122** of FIG. **2** is shown with the cover **126** removed in FIG. **3**; the cover **126** covers and protects the terminals **140** disposed in the terminal base **130**. In an embodiment, the cover **126** is a separate piece that is installed over the posts **116**. In another embodiment, the cover **126** is a potting that is filled and solidified within the end of the connector housing **122**.

The terminal base **130**, as shown in FIGS. **3** and **4**, has a plurality of terminal passageways **132** extending through the terminal base **130** in the vertical direction **V** and a hook pin **134** positioned adjacent to each of the terminal passageways **132**. Each of the hook pins **134** extends from a first end **135** adjacent to the terminal passageway **132** to a second end **136** opposite the first end **135** in a longitudinal direction **L** perpendicular to the vertical direction **V**. Each of the hook pins **134** has a protrusion **138** at the second end **136** that extends at a distance above the terminal base **130** in the vertical direction **V**. The terminal base **130** is formed of an insulative material and, in the shown embodiment, the hook pins **134** are each monolithically formed in a single piece with the terminal base **130**.

As shown in FIGS. **3** and **4**, the terminals **140** each have a termination end **141** and a mating end **142** opposite the termination end **141**. The terminals **140** each have a wire termination **144** at the termination end **141** that is capable of mechanically and electrically connecting the terminal **140** to a conductive element, such as a wire. In the shown embodiment, the wire termination **144** is a fork contact that can be welded. In other embodiments, the wire termination **144** may be any type of termination that is capable of mechanically and electrically connecting the terminal **140** to the conductive element. The terminals **140** are each formed of a conductive material and, in the shown embodiment, are monolithically formed in a single piece extending from the termination end **141** to the mating end **142**.

A process of assembling the stator **100** will now be described in greater detail.

The terminals **140** are initially positioned in the terminal passageways **132** of the terminal base **130** in a pre-bent state **P** of the terminals **140**, as shown in FIG. **4**. The termination end **141** of each of the terminals **140** protrudes beyond the terminal base **130** in the vertical direction **V**. The mating end **142** of each of the terminals **140** is exposed in the mating portion **124** of the connector interface **120**. In the shown embodiment, the stator **100** has six terminals **140** arranged in two rows. In other embodiments, the stator **100** could have any number of terminals **140** arranged in any number of rows, with a corresponding number of terminal passageways **132** and hook pins **134**.

With the terminals **140** in the pre-bent state **P** of FIG. **4**, the wires **150** are positioned in the stator **100**. The wires **150** each have a pole end **152** wrapped around one of the poles **114** of the body **110**. The pole ends **152** of the wires **150** are shown schematically around the poles **114** in FIGS. **1**, **2**, and **4** for clarity of the drawings. In an embodiment, the wires **150** are formed of a copper material.

The wires **150**, as shown in FIG. **3**, extend to a terminal end **154** opposite the pole end **152**. Between the pole end **152** and the terminal end **154**, the wires **150** are each wrapped around one of the posts **116** of the body **110**. From the post **116**, each of the wires **150** leads to one of the terminals **140**. A portion of each of the wires **150** between the post **116** and the terminal end **154** is positioned under and held by the protrusion **138** of one of the hook pins **134**. In another embodiment, the posts **116** may be omitted and the wires **150** may lead directly from the poles **114** to the protrusion **138** of one of the hook pins **134**.

The terminal end **154** of each of the wires **150** is positioned in the wire termination **144** of one of the terminals **140**, as shown in FIG. **3**. Each of the wires **150** is mechanically and electrically connected to one of the terminals **140** at the terminal end **154** by the wire termination **144**. In the shown embodiment, each of the wires **150** is welded in the wire termination **144** of one of the terminals **140**.

When the wires **150** are initially connected to the terminals **140** with the terminals **140** in the pre-bent state **P**, as shown in FIG. **3**, the wires **150** are each in a taut state **T**. In the taut state **T**, each of the wires **150** is under tension between the pole end **152** and the terminal end **154**; the wires **150** are under tension from the poles **114** to the posts **116**, from the posts **116** to the protrusions **138** of the hook pins **134**, and from the hook pins **134** to the wire terminations **144**.

From the state shown in FIG. **3**, in which the wires **150** are in the taut state **T** and the terminals **140** are in the pre-bent state **P**, the terminals **140** are bent to a bent state **B**. As shown in FIG. **5**, in an embodiment, a bending tool **300** is used to simultaneously bend a plurality of the terminals **140**. The terminals **140** are shown without the wires **150** and only with the terminal base **130** in FIG. **5** for clarity of the drawings, but the bending tool **300** is used to bend the terminals **140** with the wires **150** connected to the terminals **140** and positioned within the body **110** as described above with reference to FIG. **3**.

The bending tool **300**, as shown in the embodiment of FIG. **5**, has a pair of bending arms **310** and a bending protrusion **320** disposed at an end of each of the bending arms **310**. The bending tool **300** in the embodiment of FIG. **5** can be used to simultaneously bend two terminals **140** as described in detail below. In other embodiments, the bending tool **300** may have one bending arm **310** or more than two bending arms **310**, with a corresponding number of

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bending protrusions 320, in order to bend one or any other number of terminals 140 at a time. In each of the embodiments bending the terminals 140 described in detail below, the terminals 140 may be bent to the bent state B by moving the bending tool 300 in the longitudinal direction L or may be bent to the bent state B by any other bending device or process.

The bending of the terminals 140 shown in FIG. 3 from the pre-bent state P to the bent state B is shown in FIGS. 6A and 6B. The terminals 140 are each bent in the longitudinal direction L over the hook pin 134 that is adjacent to the terminal 140. In the bent state B shown in FIG. 6B, each of the terminals 140 has a bent section 146 at the termination end 141 that is bent to a bending angle 147 with respect to a main section 148 of the terminal 140. The bending angle 147 is greater than 0° and less than or equal to 90°. In the shown embodiment, the bending angle 147 is approximately 30°.

In the bent state B, as shown in FIG. 6B, the wire termination 144, where the terminal end 154 of the wire 150 is connected to the terminal 140, is positioned closer to the protrusion 138 of the hook pin 134 than in the pre-bent state P. The bending of the terminal 140 to the bent state B thus shortens a distance that the wire 150 extends between the protrusion 138 of the hook pin 134 and the terminal 140, allowing the wire 150 to be in a slackened state S in which the tension on the wire 150 is decreased.

A slackened portion 156 of the wire 150 is adjacent to the wire termination 144 in the bent state B, as shown in FIG. 6B, and extends at least between the wire termination 144 and the protrusion 138 of the hook pin 134. In the shown embodiment, the slackened portion 156 of the wire 150 in the slackened state S also extends between the protrusion 138 and the post 116. In all embodiments, however, the wire 150 remains taut around the poles 114 in the bent state B of the terminals 140; the slackening of the wire 150 is focused nearer to the terminals 140, in the connector interface 120.

The bending of the terminal 140 to the bent state B forms the slackened portion 156 of the wire 150 that relieves strain on the wire 150 in certain application conditions. The wire 150 with the slackened portion 156 in the slackened state S can, for example, expand and contract under rapid temperature change without increasing a tension on the wire 150 beyond a damage or breaking point. Further, the wire 150 is changed to the slackened state S without requiring an additional element of the stator 100 that must be removed after assembly; the stator 100 according to the present invention can be reliably used in a wider range of applications without requiring significant additional process steps and components, decreasing a cost of material and cost of production of the stator 100.

In the embodiment shown in FIG. 6B, the first end 135 of each of the hook pins 134 extends in the vertical direction V, perpendicular to the terminal base 130. This embodiment of the first end 135 creates the main section 148 of the terminal 140 that extends vertically above the terminal base 130, and not at the bending angle 147, when the terminal 140 is bent over the hook pin 134. In the embodiment of FIG. 6B, the terminal 140 has both the main section 138 and the bent section 146 protruding beyond the terminal base 130.

In another embodiment shown in FIG. 6C, the first end 135 of each of the hook pins 134 extends at an acute angle with respect to the vertical direction V. This embodiment of the first end 135 allows an entirety of the terminal 140 extending beyond the terminal base 130 to be bent to the bending angle 147 over the hook pin 134. In the embodiment of FIG. 6C, the main section 148 of the terminal 140 is

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disposed in the terminal base 130 and only the bent section 146 protrudes beyond the terminal base 130.

In the embodiments shown in FIGS. 3-6C, the terminals 140 each extend in a single direction beyond the terminal base 130 in the pre-bent state P; the bent section 146 is parallel to the main section 148 in the pre-bent state P. In other embodiments, shown in FIGS. 7A-8C and described in greater detail below, the terminals 140 can be formed differently in the pre-bent state P and can be bent to the bent state B to form the slackened portion 156 and the slackened state S of the wire 150.

In the embodiment shown in FIGS. 7A and 7B, the bent section 146 is perpendicular to the main section 148 in the pre-bent state P. The bent section 146 in this embodiment extends over the hook pin 134 in the longitudinal direction L in the pre-bent state P. When the terminals 140 are bent to the bent state B, as shown in FIG. 7C, the bent section 146 is further bent in the vertical direction V toward the hook pin 134 to form the bending angle 147 with respect to the main section 148, forming the slackened portion 156 and the slackened state S of the wire 150.

In another embodiment shown in FIGS. 8A and 8B, the bent section 146 is perpendicular to the main section 148 in the pre-bent state P, extending from the main section 148 in a width direction W perpendicular to the vertical direction V and the longitudinal direction L. When the terminals 140 are bent to the bent state B, as shown in FIG. 8C, the bent section 146 is bent into the longitudinal direction L to extend over the hook pin 134 to form the bending angle 147 perpendicular with respect to the main section 148, forming the slackened portion 156 and the slackened state S of the wire 150.

What is claimed is:

1. A stator, comprising:

a body having a pole;

a connector interface disposed in a portion of the body and including a terminal having a wire termination at a termination end, the terminal has a bent section at the termination end that is bent to a bending angle greater than 0° and less than or equal to 90° with respect to a main section of the terminal, the connector interface has a terminal base with a terminal passageway extending through the terminal base, the terminal is disposed in the terminal passageway and the termination end of the terminal with the bent section protrudes beyond the terminal base, the terminal base has a hook pin adjacent to the terminal in the terminal passageway, the hook pin has a protrusion at an end opposite the terminal passageway; and

a wire disposed around the pole and extending to the termination end of the terminal, the wire is electrically and mechanically connected with the terminal at the wire termination, the bent section of the terminal forms a slackened portion of the wire adjacent to the wire termination, the bent section has a pre-bent state and a bent state, the bent section has the bending angle in the bent state, the wire termination is positioned closer to the protrusion of the hook pin in the bent state than in the pre-bent state.

2. The stator of claim 1, wherein a portion of the wire between the pole and the wire termination is held by the protrusion of the hook pin.

3. The stator of claim 2, wherein the slackened portion of the wire is between the wire termination and the protrusion of the hook pin.

4. The stator of claim 1, wherein the bent section of the terminal extends over the hook pin.

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5. The stator of claim 1, wherein the bent section is parallel to the main section of the terminal in the pre-bent state.

6. The stator of claim 1, wherein the bent section is perpendicular to the main section of the terminal in the pre-bent state.

7. The stator of claim 1, wherein only the bent section of the terminal protrudes beyond the terminal base.

8. The stator of claim 1, wherein the terminal extends through the connector interface and has a mating end opposite the termination end, the terminal is monolithically formed in a single piece extending from the termination end to the mating end.

9. The stator of claim 1, wherein the pole is one of a plurality of poles of the body, the terminal is one of a plurality of terminals, and the wire is one of a plurality of wires, each of the wires is disposed around one of the poles and extends to the termination end of one of the terminals, each of the wires has the slackened portion adjacent to one of the terminals that is formed by the bent section of the one of the terminals.

10. A resolver, comprising:

a stator including a body having a pole, a connector interface disposed in a portion of the body, and a wire disposed around the pole, the connector interface includes a terminal having a termination end with a wire termination, the terminal has a bent section at the termination end that is bent to a bending angle greater than 0° and less than or equal to 90° with respect to a main section of the terminal, the connector interface has a terminal base with a terminal passageway extending through the terminal base, the terminal is disposed in the terminal passageway and the termination end of the terminal with the bent section protrudes beyond the terminal base, the terminal base has a hook pin adjacent to the terminal in the terminal passageway, the hook pin has a protrusion at an end opposite the terminal passageway, the wire extends to the termination end of the terminal and is electrically and mechanically connected with the terminal at the wire termination, the bent section of the terminal forms a slackened portion of the wire adjacent to the wire termination, the bent section has a pre-bent state and a bent state, the bent section has

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the bending angle in the bent state, the wire termination is positioned closer to the protrusion of the hook pin in the bent state than in the pre-bent state.

11. The resolver of claim 10, wherein the body is an approximately circular member defining a rotor receiving space, and further comprising a rotor disposed in the rotor receiving space, the rotor is rotatable with respect to the stator.

12. A method of forming a slackened portion of a wire in a stator, comprising:

providing a stator having a body with a pole and a connector interface disposed in a portion of the body, the connector interface includes a terminal having a wire termination at a termination end;

positioning the wire around the pole and extending to the termination end of the terminal, the wire is electrically and mechanically connected with the terminal at the wire termination with the wire in a taut state and the terminal in a pre-bent state; and

bending the terminal from the pre-bent state to a bent state, the bending of the terminal to the bent state slackens the wire to a slackened state in which the wire has a slackened portion adjacent to the wire termination.

13. The method of claim 12, wherein a bending tool is used to bend the terminal to the bent state in the bending step.

14. The method of claim 12, wherein the bending step bends a bent section of the terminal to a bending angle greater than 0° and less than or equal to 90° with respect to a main section of the terminal.

15. The method of claim 12, wherein the connector interface has a terminal base with a terminal passageway extending through the terminal base and a hook pin adjacent to the terminal passageway, the hook pin has a protrusion at an end opposite the terminal passageway.

16. The method of claim 15, wherein the terminal is bent in the bending step in a direction toward the protrusion and over the hook pin.

17. The method of claim 15, wherein the bending step positions the wire termination closer to the protrusion of the hook pin to form the slackened portion of the wire.

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